# Benjamin Dixon

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## Education

**University of Edinburgh** — Edinburgh, UK
*MSc Cognitive Science* — *2026 – 2027*

**Emory University** — Atlanta, GA, USA
*BA Computer Science; BA Psychology & Linguistics* — *2021 – 2025*

Graduated with Highest Honors (honors thesis); Highest Excellence in Psychology & Linguistics

Thesis: *Automated Assessment of Concrete Language in Clinical High-Risk for Psychosis: A Novel Large Language Model Approach*

## Research Interests

Computational psychiatry; neural and behavioral signatures of prediction in language; temporal structure of speech and neural time series; information-theoretic approaches to language; assistive technology for disabled populations

## Publications & Presentations

### Peer-Reviewed Conference Papers

Oliveira, G. C., **Dixon, B. T.**, et al. (2026). *DIALOG-DeID: Role- and Privacy-Aware Transcription for Clinical Interviews Beyond WER.* In *Proceedings of Interspeech 2026*, Sydney, Australia.

### Manuscripts in Preparation

**Dixon, B. T.**, et al. (in preparation). *Measuring Cohesion with Self-Attention: Local Focus and Weaker Long-Range Links in Clinical High Risk for Psychosis.* Invited submission, *Schizophrenia Bulletin* special issue (AMP SCZ baseline analyses).

Bilgrami, Z. R., **Dixon, B. T.**, et al. (in preparation). *Semantic Elements of Interview and Diary Speech Characterize Positive and Negative Symptoms in Clinical High Risk for Psychosis Across Languages: Baseline Findings from the Accelerating Medicines Partnership Schizophrenia (AMP SCZ) Observational Study.* Invited submission, *Schizophrenia Bulletin* special issue (AMP SCZ baseline analyses).

### Conference Posters

Bilgrami, Z. R., **Dixon, B. T.**, et al. (2027, January). *Large Language Models Can Predict Psychosis Risk from Disruptions in the Flow of Ideas: Results from the AMP SCZ Observational Study.* Accepted poster, American College of Neuropsychopharmacology (ACNP) 2027 Annual Meeting, San Diego, CA, USA.

**Dixon, B. T.**, et al. (2026, March). *Altered Pronoun Use in Clinical High Risk for Psychosis Across Languages.* Schizophrenia International Research Society (SIRS) 2026 Congress, Florence, Italy.

Bilgrami, A., **Dixon, B. T.**, et al. (2026, March). *Uncovering Universal and Culture-Specific Themes in Psychosis Risk: A Cross-Linguistic Large Language Model Framework.* SIRS 2026 Congress, Florence, Italy.

Zhang, Y., **Dixon, B. T.**, et al. (2026, March). *Reduced Future Orientation and Temporal Switching in Spontaneous Language at Clinical High Risk for Psychosis.* SIRS 2026 Congress, Florence, Italy.

Zarina, B., **Dixon, B. T.**, Wolff, P. (2025, April). *Automated Assessment of Language Concreteness in Clinical High-Risk and Healthy Control Speech Using Large Language Models.* SIRS 2025 Congress, Chicago, IL, USA.

### Submitted Abstracts

Tamayo, Z., **Dixon, B. T.**, et al. (submitted). *A Scalable and Extensible Infrastructure for AI-Driven Processing of Clinical Interviews.* ACNP 2027 Annual Meeting, San Diego, CA, USA.

### Other Presentations

**Dixon, B. T.**, Sabrina, S., Segall, M. (2024, October). *Developing Autistic Student Leadership.* College Autism Summit, Columbus, OH, USA.

## Research Experience

**Language as a Biomarker Lab, Emory University** — Atlanta, GA, USA

*Consulting Scientist* — *Sep 2026 – Present*

- Model clinical interviews as **time series** of sentence embeddings and word-level **surprisal**, using **detrended fluctuation analysis** to test whether the long-range temporal structure of speech differs in early **psychosis** (**NumPy**, **scikit-learn**, **Polars**).
- Built attention-pooling **neural network** classifiers (**PyTorch**) that combine sentence-level LLM **embeddings** into one representation per interview, separating clinical high-risk participants from healthy controls (**cross-validation**, **AUROC**).
- Fine-tune LLMs on naturalistic dialogue so that model-based measures of word predictability match human predictability judgments (**model fine-tuning**, **behavioral data**).
- Maintain **GCP** cloud infrastructure; deployed a local LLM assistant on the lab **HPC** cluster for non-technical staff.

*Research Software Engineer* — *Mar 2025 – Aug 2026*

- Led design of the speech and language feature set released to the NIMH Data Archive (Release 4) for the AMP SCZ consortium, covering 15,000+ interview transcripts in 6 languages and giving external researchers standardized **biomarkers** of psychosis risk.
- Compared clinical-high-risk and healthy-control groups with **Bayesian mixed-effects models** (brms) in **R**, highlighting group variability.
- Trained linear **decoding models** (ridge, logistic regression) and shallow neural networks on LLM embeddings (Llama 3.3, gpt-oss-120b, DeBERTa) to predict human ratings of concreteness and specificity from transcripts.
- Built cross-linguistic theme discovery with GPU-accelerated **clustering** (HDBSCAN) of sentence embeddings, replacing a method that did not scale to the full dataset.
- Built automated speech-processing pipelines (speaker separation with NVIDIA NeMo and pyannote; transcription with Whisper and Parakeet) on a GPU **HPC cluster**, halving inference time through quantization and batching; processed 1,500 hours of interview audio.
- Supervised three undergraduate research assistants.

*Research Assistant* — *May 2024 – Mar 2025*

- Measured concreteness of speech with **large language models** in clinical high-risk and healthy participants; results presented at SIRS 2025 and in an honors thesis awarded Highest Honors.

**Directed Research, Emory University** — Atlanta, GA, USA

*Undergraduate Researcher* — *Sep 2023 – May 2024*

- Built a question-answering system over technical documentation using retrieval-augmented generation (**RAG**, Haystack) and a fine-tuned Flan-T5-Large model, reducing fabricated answers.

## Projects

**Word Surprisal and EEG Encoding Models** — Oct 2026

*Independent Project* — GitHub

- Tested whether language-model **surprisal** predicts **EEG** from 19 people listening to an audiobook, fitting ridge **encoding models** (temporal response functions, **MNE-Python**) with leave-one-run-out cross-validation.
- Built the **signal processing** pipeline (re-referencing, 0.5–8 Hz band-pass filtering, bad-channel interpolation) and aligned EEG with word timings from a separate transcript source.
- Surprisal improved prediction in 18 of 19 participants ($p < 0.001$) beyond sound, word-timing, and word-frequency controls; a **cluster permutation test** localized the effect to a centro-parietal response at about 400 ms (N400).

**LifeLine: Plain-Language Guide to Edinburgh Public Services** — Oct 2026

*1st Place, UN Sustainable Development Goals Challenge, AdaHack 2026* — GitHub

- Fine-tuned a 1.5B-parameter language model (Qwen2.5, **LoRA**) to rewrite service descriptions in EasyRead plain language for readers with learning disabilities, running offline on a laptop CPU or GPU; team of 3.

**Mammography + Chest X-Ray Embedding Fusion** — Aug 2026

*Team Lead, HITI Lab Datathon* — GitHub

- Collaborated with an 8-person team of engineers and clinicians testing whether combining two imaging models' embeddings predicts 4 chronic diseases better than either alone; fusion improved diabetes and kidney disease screening.
- Compared penalized logistic regression and early-, late-, and gated-fusion neural networks under **nested cross-validation** (**PyTorch**, **scikit-learn**), and evaluated fairness across age, race, and BMI subgroups.

## Teaching & Mentoring

**Math & Computer Science Lab Assistant, Emory University** — Atlanta, GA, USA
*Tutor and Technical Support* — *Sep 2023 – Jan 2024*

- Tutored introductory computer science students in **data structures and algorithms**; supported students and faculty on **macOS**, **Windows**, and **Linux**.

**Tech Academy** — Little Canada, MN, USA
*Summer Camp Instructor* — *Summer 2022*

- Designed and taught Python and robotics lessons for schoolchildren, adapting material to different learning needs.

## Activities & Service

**HumanEd, University of Edinburgh** — Edinburgh, UK
*Advanced Team Member, Computer Vision* — *2026 – Present*

**Common Ground Leadership Programme, University of Edinburgh** — Edinburgh, UK
*Inaugural Cohort (Edinburgh Award)* — *2026 – 2027*

**Emory Autism Advocacy Organization** — Atlanta, GA, USA
*President; Autism Celebration Month Committee Chair* — *2023 – 2025*

- Grew membership by over 50% and organized three annual Autism Celebration Months, including a research symposium featuring neurodivergent researchers.
- Co-led a campus sensory-friendly spaces initiative with university administration and facilities management.

## Honors & Awards

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| 1st Place, UN Sustainable Development Goals Challenge, AdaHack 2026, University of Edinburgh | 2026 |
| Highest Honors, Emory College of Arts and Sciences (honors thesis) | 2025 |
| Highest Excellence in Psychology & Linguistics, Emory University | 2025 |
| Psi Chi International Honor Society in Psychology | 2024 |

## Skills

**Methods**: EEG analysis, signal processing, time series analysis, encoding/decoding models, Bayesian and mixed-effects modeling, machine learning, clustering, NLP, LLM fine-tuning
**Programming**: Python (MNE, NumPy, pandas, Polars, scikit-learn, PyTorch, HuggingFace, vLLM), R (brms, GAMLSS), SQL, Bash, Java, C/C++, JavaScript
**Infrastructure**: HPC, Slurm, GPU computing, Docker, Git, GCP, AWS (SageMaker, S3)